

CS 2601 Linear and Convex Optimization

14. Dual LP

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Outline

- Dual LP
- Interpretation of dual problems
- Weak and strong duality
- Dual LP via Lagrangian

Lower bounds in LP

$$\begin{array}{ll}\min_{\mathbf{x} \in \mathbb{R}^2} & f(\mathbf{x}) = x_1 + 2x_2 \\ \text{s.t.} & 2x_1 + x_2 \geq 2 \\ & x_1, x_2 \geq 0\end{array}$$

Any feasible solution $\mathbf{x}_0$ gives an upper bound on the optimal value f^* ,

$$f^* \leq f(\mathbf{x}_0)$$

Can we also get a lower bound f_{LB} on f^* ?

$$f^* \geq f_{\text{LB}}$$

Note. A lower bound on f^* is the same as a lower bound on $f(\mathbf{x})$ for all feasible $\mathbf{x} \in X$, i.e.

$$f^* \geq f_{\text{LB}} \iff f(\mathbf{x}) \geq f_{\text{LB}}, \quad \forall \mathbf{x} \in X$$

Lower bounds in LP (cont'd)

For any $\mu_1, \mu_2, \mu_3 \geq 0$,

$$\begin{array}{rcll} \mu_1 \times & [& 2x_1 & + & x_2 & \geq & 2 &] \\ \mu_2 \times & [& x_1 & & & \geq & 0 &] \\ \mu_3 \times & [& & & x_2 & \geq & 0 &] \\ \hline & (2\mu_1 + \mu_2)x_1 & + & (\mu_1 + \mu_3)x_2 & \geq & 2\mu_1 =: \psi(\boldsymbol{\mu}) \end{array}$$

We can set $2\mu_1 + \mu_2 = 1$ and $\mu_1 + \mu_3 = 2$ so the LHS becomes f .

Thus

$$f(\mathbf{x}) \geq \psi(\boldsymbol{\mu}) = 2\mu_1$$

for any $\mathbf{x} \in X$ and any μ_1, μ_2, μ_3 s.t.

$$2\mu_1 + \mu_2 = 1, \quad \mu_1 + \mu_3 = 2, \quad \mu_1, \mu_2, \mu_3 \geq 0$$

In particular, $f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}) \geq \psi(\boldsymbol{\mu})$ for such $\boldsymbol{\mu}$.

Lower bounds in LP (cont'd)

The quality of the lower bound $\psi(\mu) = 2\mu_1$ varies for different μ .

- $\psi(0, 1, 2) = 0$, so

$$0 = \psi(0, 1, 2) \leq f^* \leq f(1, 0) = 1$$

which also tells us $x_0 = (1, 0)$ is 1-suboptimal, i.e.

$$f(1, 0) - f^* \leq 1$$

- $\psi(\frac{1}{2}, 0, \frac{3}{2}) = 1$, so

$$1 = \psi(\frac{1}{2}, 0, \frac{3}{2}) \leq f^* \leq f(1, 0) = 1$$

which tells us $f^* = 1$ and $x_0 = (1, 0)$ is the optimal solution!

Dual LP

To get the best lower bound, we maximize over μ_1, μ_2, μ_3 ,

$$\min_{\mathbf{x} \in \mathbb{R}^2} \quad f(\mathbf{x}) = x_1 + 2x_2$$

$$\text{s.t.} \quad 2x_1 + x_2 \geq 2$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

primal LP

$$\max_{\boldsymbol{\mu} \in \mathbb{R}^3} \quad \psi(\boldsymbol{\mu}) = 2\mu_1$$

$$\text{s.t.} \quad 2\mu_1 + \mu_2 = 1$$

$$\mu_1 + \mu_3 = 2$$

$$\mu_1 \geq 0$$

$$\mu_2 \geq 0$$

$$\mu_3 \geq 0$$

dual LP

The variables μ_1, μ_2, μ_3 are called **dual variables**.

We have one dual variable for each constraint in the primal problem.

Dual LP (cont'd)

Given $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{k \times n}$, $\mathbf{b} \in \mathbb{R}^k$, $\mathbf{G} \in \mathbb{R}^{m \times n}$, $\mathbf{h} \in \mathbb{R}^m$, consider

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{Gx} \geq \mathbf{h} \end{aligned}$$

For $\boldsymbol{\lambda} \in \mathbb{R}^k$, $\boldsymbol{\mu} \in \mathbb{R}^m$ and $\boldsymbol{\mu} \geq \mathbf{0}$,

$$\boldsymbol{\lambda}^T \mathbf{Ax} + \boldsymbol{\mu}^T \mathbf{Gx} \geq \boldsymbol{\lambda}^T \mathbf{b} + \boldsymbol{\mu}^T \mathbf{h} =: \psi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

If $\mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c}$, then we can lower bound f^* by $f^* \geq \psi(\boldsymbol{\lambda}, \boldsymbol{\mu})$.

To maximize the lower bound, solve the following dual LP

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \psi(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathbf{b}^T \boldsymbol{\lambda} + \mathbf{h}^T \boldsymbol{\mu} \\ \text{s.t.} \quad & \mathbf{A}^T \boldsymbol{\lambda} + \mathbf{G}^T \boldsymbol{\mu} = \mathbf{c} \\ & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

Dual LP (cont'd)

It is common to eliminate dual variables corresponding to $x \geq 0$, and call the result the dual LP. Here are some common forms of dual LP,

$$\begin{array}{ll}\min_x & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

$$\begin{array}{ll}\max_y & \mathbf{b}^T \mathbf{y} \\ \text{s.t.} & \mathbf{A}^T \mathbf{y} \leq \mathbf{c}\end{array}$$

$$\begin{array}{ll}\min_x & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \geq \mathbf{b}\end{array}$$

$$\begin{array}{ll}\max_y & \mathbf{b}^T \mathbf{y} \\ \text{s.t.} & \mathbf{A}^T \mathbf{y} = \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}\end{array}$$

$$\begin{array}{ll}\min_x & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{Ax} \geq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

$$\begin{array}{ll}\max_y & \mathbf{b}^T \mathbf{y} \\ \text{s.t.} & \mathbf{A}^T \mathbf{y} \leq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0}\end{array}$$

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Manufacturing problem

Suppose company A can manufacture n products from m materials.

- Manufacturing one unit of product i needs a_{ij} units of material j
- The unit price of product i is b_i
- The inventory of material j is c_j

where $a_{ij} > 0$, $b_i > 0$, $c_j > 0$.

Let y_i denote the amount of product i manufactured. To maximize its revenue, the company solves the following LP,

$$\begin{aligned} \max_{\mathbf{y}} \quad & \sum_{i=1}^n b_i y_i \\ \text{s.t.} \quad & \sum_{i=1}^n y_i a_{ij} \leq c_j, \forall j = 1, 2, \dots, m \\ & y_i \geq 0, \quad i = 1, 2, \dots, n \end{aligned}$$

Manufacturing problem (cont'd)

Now suppose company B offers to buy the raw materials at the price of x_j per unit of material j .

The equivalent offer for one unit of product i is $\sum_{j=1}^m a_{ij}x_j$. Company A will accept the offer only if

$$\sum_{j=1}^m a_{ij}x_j \geq b_i, \quad i = 1, 2, \dots, n$$

To minimize its cost, company B solves the dual LP,

$$\begin{aligned} \min_{\mathbf{x}} \quad & \sum_{j=1}^m c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^m a_{ij} x_j \geq b_i, \forall i = 1, 2, \dots, n \\ & x_j \geq 0, \quad j = 1, 2, \dots, m \end{aligned}$$

Optimal transport problem

Recall the optimal transport problem in §1,

$$\begin{aligned} \min_{(x_{ij})} \quad & \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij} \\ \text{s. t.} \quad & \sum_{i=1}^n x_{ij} = b_j \quad \text{for } j = 1, 2, \dots, m \\ & \sum_{j=1}^m x_{ij} \leq a_i \quad \text{for } i = 1, 2, \dots, n \\ & x_{ij} \geq 0 \quad \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, m \end{aligned}$$

- x_{ij} : quantity shipped from warehouse i to customer j
- c_{ij} : unit shipping cost from warehouse i to customer j
- a_i : inventory at warehouse i
- b_j : demand at customer j

Optimal transport problem (cont'd)

Now instead of actually shipping the products, you decide to fulfill the orders by trading with another seller of the same product, who

- buys your stock at warehouse i at the unit price μ_i
- delivers the order of customer j at the unit price λ_j

The cost of “sending” one unit from i to j is $\lambda_j - \mu_i$. The deal will be competitive if

$$\lambda_j - \mu_i \leq c_{ij}$$

To maximize his profit, the other seller solves the dual LP,

$$\begin{array}{ll} \max_{(\lambda_j), (\mu_i)} & \sum_{j=1}^m b_j \lambda_j - \sum_{i=1}^n a_i \mu_i \\ \text{s. t.} & \lambda_j - \mu_i \leq c_{ij} \quad \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, m \\ & \mu_i \geq 0 \quad \text{for } i = 1, 2, \dots, n \end{array}$$

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Weak and strong duality

Consider an LP and its dual. WLOG, consider the inequality form,

$$\begin{array}{ll} \min_x & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} & \mathbf{A}\mathbf{x} \geq \mathbf{b} \end{array} \quad (\text{P}) \quad \left| \quad \begin{array}{ll} \max_y & \mathbf{b}^T \mathbf{y} \\ \text{s.t.} & \mathbf{A}^T \mathbf{y} = \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0} \end{array} \quad (\text{D})$$

Weak duality. If $\mathbf{x}$ is primal feasible and $\mathbf{y}$ is dual feasible, then

$$\mathbf{c}^T \mathbf{x} \geq \mathbf{b}^T \mathbf{y}$$

Proof.

$$\mathbf{c}^T \mathbf{x} = (\mathbf{A}^T \mathbf{y})^T \mathbf{x} = \mathbf{y}^T \mathbf{A} \mathbf{x} \geq \mathbf{y}^T \mathbf{b}$$

Strong duality. If either (P) or (D) has a finite optimal value, so does the other, the optimal solutions $\mathbf{x}^*$ and $\mathbf{y}^*$ exist, and $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$.

Proof. Will follow from Slater's Theorem. Show weaker form next.

Strong duality

Lemma. If all (equality and inequality) constraint functions are affine, then the KKT conditions hold at a local minimum.

Note. We do not assume the local minimum is a regular point.

Theorem. If (P) has an optimal solution, then (D) also has an optimal solution and strong duality holds.

Proof. Let $\mathbf{x}^*$ be the optimal solution of (P). By the KKT conditions, there exist Lagrange multipliers $\boldsymbol{\mu}^*$ s.t.

1. $\mathbf{c} - \mathbf{A}^T \boldsymbol{\mu}^* = \mathbf{0}$ (stationarity)
2. $\boldsymbol{\mu}^* \geq \mathbf{0}$ (nonnegativity)
3. $(\boldsymbol{\mu}^*)^T (\mathbf{A} \mathbf{x}^* - \mathbf{b}) = 0$ (complementary slackness)

By 1 and 2, $\boldsymbol{\mu}^*$ is feasible for (D). By 1 and 3,

$$\mathbf{c}^T \mathbf{x}^* = (\mathbf{A} \boldsymbol{\mu}^*)^T \mathbf{x}^* = \mathbf{b}^T \boldsymbol{\mu}^*$$

By weak duality, $\boldsymbol{\mu}^*$ is optimal for (D) and hence strong duality holds.

Example

Recall the following pair of primal and dual LP,

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) = x_1 + 2x_2$$

$$\text{s.t. } 2x_1 + x_2 \geq 2$$

$$\mathbf{x} \geq \mathbf{0}$$

$$\max_{\boldsymbol{\mu} \in \mathbb{R}^3} \psi(\boldsymbol{\mu}) = 2\mu_1$$

$$\text{s.t. } 2\mu_1 + \mu_2 = 1$$

$$\mu_1 + \mu_3 = 2$$

$$\boldsymbol{\mu} \geq \mathbf{0}$$

- The primal LP can be solved graphically, $\mathbf{x}^* = (1, 0)$, $f^* = 1$.
- The dual is equivalent to

$$\max_{\mu_1} 2\mu_1$$

$$\text{s.t. } 2\mu_1 \leq 1$$

$$\mu_1 \leq 2$$

$$\mu_1 \geq 0$$

So $\boldsymbol{\mu}^* = (\frac{1}{2}, 0, \frac{3}{2})$, $\psi^* = 1 = f^*$.

Wasserstein distance

In the optimal transport problem, let $\mathbf{x}_i$ be the location of warehouse i , and $\mathbf{y}_j$ be the location of customer j . Suppose

$$\sum_{i=1}^n a_i = 1 = \sum_{j=1}^m b_j$$

Note a, b can be interpreted as two discrete probability distributions,

$$\Pr[\mathbf{X} = \mathbf{x}_i] = a_i, \quad \Pr[\mathbf{Y} = \mathbf{y}_j] = b_j$$

Now suppose the shipping cost c_{ij} is determined by the distance between $\mathbf{x}_i$ and $\mathbf{y}_j$,

$$c_{ij} = d(\mathbf{x}_i, \mathbf{y}_j)$$

e.g. $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2$.

Wasserstein distance (cont'd)

The (first) Wasserstein distance $W_1(a, b)$ between the two distributions a and b is the optimal value of the optimal transport problem.

$$\begin{aligned} \min_{(x_{ij})} \quad & \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij} \\ \text{s. t.} \quad & \sum_{i=1}^n x_{ij} = b_j \quad \text{for } j = 1, 2, \dots, m \\ & \sum_{j=1}^m x_{ij} = a_i \quad \text{for } i = 1, 2, \dots, n \\ & x_{ij} \geq 0 \quad \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, m \end{aligned}$$

Note the feasible set is $\Pi(a, b)$, the set of all joint distributions with marginals a and b ,

$$W_1(a, b) = \inf_{x \in \Pi(a, b)} \sum_{i,j} d(\mathbf{x}_i, \mathbf{y}_j) x_{ij} = \inf_{x \in \Pi(a, b)} \mathbb{E}_{(\mathbf{X}, \mathbf{Y}) \sim x} [d(\mathbf{X}, \mathbf{Y})]$$

Wasserstein distance (cont'd)

The dual problem is

$$\begin{aligned} \max_{(\lambda_j), (\mu_i)} \quad & \sum_{j=1}^m b_j \lambda_j - \sum_{i=1}^n a_i \mu_i \\ \text{s. t.} \quad & \lambda_j - \mu_i \leq c_{ij} \quad \text{for } i = 1, 2, \dots, n; j = 1, 2, \dots, m \end{aligned}$$

Note. We do not have the constraints $\mu_i \geq 0$ as on slide 12 due to the modification in the primal, “ $\leq a_i$ ” $\rightarrow$ “ $= a_i$ ”.

By strong duality, $W_1(a, b)$ is also equal to the dual optimal value.

- $x_{ij} = a_i b_j$ is primal feasible (what's the probabilistic interpretation?)
- the optimal cost is lower bounded by $\min_{i,j} c_{ij}$ and hence finite.

Rewriting $\lambda_j = \lambda(\mathbf{y}_j)$ and $\mu_i = \mu(\mathbf{x}_i)$,

$$W_1(a, b) = \sup_{\lambda(\mathbf{y}_j) - \mu(\mathbf{x}_i) \leq d(\mathbf{x}_i, \mathbf{y}_j), \forall i,j} \left[\sum_{j=1}^m b_j \lambda(\mathbf{y}_j) - \sum_{i=1}^n a_i \mu(\mathbf{x}_i) \right]$$

Wasserstein distance (cont'd)

We can actually restrict λ and μ to be the same function. Then the constraints become

$$h(\mathbf{y}_j) - h(\mathbf{x}_i) \leq d(\mathbf{x}_i, \mathbf{y}_j), \quad \forall i, j$$

which can be further strengthened to 1-Lipschitz continuity of h , i.e.

$$h(\mathbf{z}) - h(\mathbf{z}') \leq d(\mathbf{z}, \mathbf{z}'), \quad \forall \mathbf{z}, \mathbf{z}'$$

Theorem (Kantorovich-Rubinstein duality).

$$W_1(a, b) = \sup_{h \text{ is 1-Lipschitz}} \left[\sum_{j=1}^m b_j h(\mathbf{y}_j) - \sum_{i=1}^n a_i h(\mathbf{x}_i) \right]$$

In the general case,

$$\begin{aligned} W_1(a, b) &= \inf_{x \in \Pi(a, b)} \mathbb{E}_{(\mathbf{X}, \mathbf{Y}) \sim x} [d(\mathbf{X}, \mathbf{Y})] \\ &= \sup_{h \text{ is 1-Lipschitz}} \mathbb{E}_{\mathbf{Y} \sim b} [h(\mathbf{Y})] - \mathbb{E}_{\mathbf{X} \sim a} [h(\mathbf{X})] \end{aligned}$$

Appendix: Proof of discrete case

Given μ , define

$$h(\mathbf{z}) = \min_k [\mu(\mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{z})]$$

Then h is 1-Lipschitz. Indeed, by the triangle inequality for d ,

$$\begin{aligned} h(\mathbf{z}) &= \min_k [\mu(\mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{z})] \\ &\leq \min_k \{\mu(\mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{z}') + d(\mathbf{z}, \mathbf{z}')\} = h(\mathbf{z}') + d(\mathbf{z}, \mathbf{z}') \end{aligned}$$

If λ and μ satisfy the dual constraints, then replacing both λ and μ by h yields a solution that is at least as good as the old one,

- feasibility follows from 1-Lipschitz continuity of h ,

$$h(\mathbf{y}_j) - h(\mathbf{x}_i) \leq d(\mathbf{x}_i, \mathbf{y}_j), \quad \forall i, j$$

- improvement follows from the dual constraints and the definition of h ,

$$\lambda(\mathbf{y}_j) \leq \min_k [\mu(\mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{y}_j)] = h(\mathbf{y}_j), \quad \forall j$$

$$\mu(\mathbf{x}_i) = \mu(\mathbf{x}_i) + d(\mathbf{x}_i, \mathbf{x}_i) \geq \min_k [\mu(\mathbf{x}_k) + d(\mathbf{x}_k, \mathbf{x}_i)] = h(\mathbf{x}_i), \quad \forall i$$

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Dual LP via Lagrangian

The Lagrangian for the general LP on slide 6 is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = \mathbf{c}^T \mathbf{x} - \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b}) - \boldsymbol{\mu}^T (\mathbf{G}\mathbf{x} - \mathbf{h})$$

If $\boldsymbol{\mu} \geq \mathbf{0}$ and $\mathbf{x} \in X$, i.e. $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{G}\mathbf{x} \geq \mathbf{h}$, then

$$f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} \geq \underbrace{\mathbf{c}^T \mathbf{x} - \boldsymbol{\lambda}^T (\mathbf{A}\mathbf{x} - \mathbf{b})}_{=0} - \underbrace{\boldsymbol{\mu}^T (\mathbf{G}\mathbf{x} - \mathbf{h})}_{\geq 0} = \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$$

Taking the infimum over $\mathbf{x} \in X$ first and then relaxing the constraint,

$$f^* = \inf_{\mathbf{x} \in X} f(\mathbf{x}) \geq \inf_{\mathbf{x} \in X} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \geq \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) =: \phi(\boldsymbol{\lambda}, \boldsymbol{\mu})$$

To maximize the lower bound, solve the following dual problem

$$\begin{aligned} \max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \quad & \phi(\boldsymbol{\lambda}, \boldsymbol{\mu}) \\ \text{s.t.} \quad & \boldsymbol{\mu} \geq \mathbf{0} \end{aligned}$$

Dual LP via Lagrangian (cont'd)

Note

$$\mathcal{L}(x, \lambda, \mu) = (c - A^T \lambda - G^T \mu)^T x + b^T \lambda + h^T \mu.$$

An affine function is bounded below iff the coefficient for x is zero¹.

The dual problem

$$\begin{aligned} \max_{\lambda, \mu} \quad \phi(\lambda, \mu) &= \begin{cases} b^T \lambda + h^T \mu, & \text{if } c - A^T \lambda - G^T \mu = 0 \\ -\infty & \text{otherwise} \end{cases} \\ \text{s.t.} \quad \mu &\geq 0 \end{aligned}$$

which is equivalent to the dual LP

$$\begin{aligned} \max_{\lambda, \mu} \quad \psi(\lambda, \mu) &= b^T \lambda + h^T \mu \\ \text{s.t.} \quad A^T \lambda + G^T \mu &= c \\ \mu &\geq 0 \end{aligned}$$

¹Consider $f(x) = a^T x + c$. If $a = 0$, then $\inf_x f(x) = c$. If $a \neq 0$, letting $x = -ta$ and $t \rightarrow +\infty$ yields $\inf_x f(x) \leq -t\|a\|^2 + c \rightarrow -\infty$.